THE UNITED REPUBLIC OF TANZANIA
PRESIDENT'S OFFICE
REGIONAL ADMINISTRATION AND LOCAL GOVERNMENT

IRINGA, LINDI AND MOROGORO FORM FOUR REGIONAL MOCK EXAMINATIONS, 2024
BIOLOGY 1

CODE: 033/1
TIME: 3 HOURS

May, 2024

INSTRUCTIONS

1. This paper consists of sections **A, B** and **C** with a total of **eleven (11)** questions.
2. Answer **all** questions in the sections **A** and **B** and two **(2)** questions from section **C.**
3. Section A carries **sixteen (16)** marks, section **(B) fifty-four (54)** marks and section **C** carries **thirty (30)** marks.
4. All writing should be in **blue** or **black** ink, except for diagrams that must be drawn in pencil.
5. Communication devices and any unauthorized materials are **not** allowed in the examination room.
6. Remember to write your candidate's examination number at the top of every page.

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SECTION A (16 MARKS)

Answer all questions in this section

1. For each of the items (i) - (x), choose the correct answer from among the given alternatives and write its letter beside the item number in the answer sheet provided.

i. Form two students were observing the structures of animal cells by electron microscope in the biology laboratory. Unfortunately, the image formed from their microscopes was not clear. Which of the following part of microscope will they use to adjust in order to obtain a sharp image of animal cells?

A. Rotating nose piece
B. Coarse adjustment knob
C. Fine adjustment knob
D. Eyepiece lens
E. Mirror

ii. One of the following is the correct nitrogenous base pairing for DNA molecules?

A. Uracil{U} and adenine{A}
B. Adenine {A}and guanine {G}
C. Thyamine{T}and cytocine{C}
D. Guanine{G} and cytocine {C}

iii. The small bean shaped glands which filter lymph and damaged cells are found in the following parts of the body except;

A. Abdomen
B. Groin
C. Armpit
D. Brain
E. Neck

iv. During the graduation ceremony, a certain student ate meat, as a form four student state where that food was first digested in the digestive system?

A. Mouth
B. Oesophagus
C. Stomach
D. Small intestine
E. Duodenum

v. After diagnosis of an accident victim, doctor said 'his radius and ulna are fractured; which part of the body was he talking about?

A. Bones of the skull
B. Shoulder
C. Forearm
D. Bones of the foot
E. Bones of the upper arm

vi. The type of cell division that occur in somatic cells leading to growth and development of a living organism. How many identical daughter cells are produced during this process

A. Four haploid
B. Four diploid
C. One haploid
D. Two haploid
E. Two diploid

vii. During the biology class session, a biology teacher explained to his students about a theory of origin of life which state that "the universe has always existed and has no origin, that is, it did not have a moment of creation thus life has no origin." Which of the following is the correct name of this theory?

A. The steady state theory
B. Special creation theory
C. Organic evolution theory
D. Cosmozoan theory
E. Spontaneous generation theory

viii. Most of students have failed to make a clear distinction between members of class arachnida from those of class insecta when they were classifying different living organisms in biology laboratory. As the expert of biology, which characteristic distinguish the two groups of phylum arthropoda?

A. The insects have three body parts while arachnids have four body parts
B. The insects have one pair of antennae while arachnids have two pair of antenna
C. Insects have six pair of walking legs while arachnids have three pair of walking legs
D. Insects are carnivorous while arachnids are herbivorous
E. Insects carry out gaseous exchange through trachea system while arachnids carry out gaseous exchange through book lungs

ix. During ultrafiltration in the glomerulus too large materials are retained in the blood capillaries while the smaller ones are filtered into the Bowman's capsule. The filtered material includes

A. Glucose molecules, water, urea and dissolved mineral salts
B. Cells, proteins and glucose
C. Uric acids, proteins and urea
D. Red blood cells, urea and water
E. Proteins, water, urea, fatty acids, and glucose

x. The emerging root when the seed start to germinate is known as

A. Cotyledone
B. Micropyle
C. Plumule
D. Radicle
E. Coleorhiza

2. Match the items in List A with the responses in List B by writing the letter of the correct responses in List B besides the item number in the answer booklet provided

COLUMN A	COLUMN B
i. These organisms reproduce sexually through formation of zygospores.	A Ferns.
ii These organisms reproduce sexually through formation of basidia.	B Maize plants.
iii These organisms have spores producing organ called sori.	C Mosses.
iv These organisms have ovules within the ovary in which after fertilization ovules and ovary become seeds and fruits respectively.	D Mucors.
v. These organisms reproduce sexually through formation of ascospores.	E Mushrooms.
vi These organisms reproduce sexually through cones	F Pine trees.
	G Virus.
	H Yeasts.

SECTION B: (54 Marks).

Answer all questions in this section

3. a) A laboratory technician of Mavanga secondary school, he arranged the apparatus listed below on the laboratory working benches in order for his students of form four class to conduct a food test experiment. Assume you are the one that you are going to use these apparatus during food test experiment. Briefly explain how will you use them correctly?

i. Test tube
ii. Motor and pestle
iii. Beaker
iv. Test tube rack
v. Dropper

b) In schools, laboratories are quite different from other school facilities. What are scientific reasons for having the following laboratory features?

i. Large windows
ii. Water
iii. Heat source
iv. Emergency door

4. a) Plant growth can be limited by many factors. Explain internal factors only, limiting your answer to three points.
   (b) Adolescence is an important group in the community, which signifies that a person is in transition stage to adulthood, so some services are needed so as to make them live fairly in their environment. Using three points explain those services.
5. Juma was narrating a story in a class about a person who has unbelievable strength and energy when subjected in an extremely dangerously situation.
   i. What is the chemical substance released in Juma's body during that situation?
   ii. What are the five roles of the chemical substance mentioned in 5(i) above?

6. Explain the following phenomena as used in biology.
   i. Insulin and glucagon hormones work antagonistically
   ii. In cold and humidity weather, human usually produce large quantity of urine that is colorless.
   iii. Fish dies when taken outside water bodies
   iv. When plant seedling placed horizontally on ground for some days, the root grow downward while the shoot grow upward
   v. It is difficult to detect smell of substances when we are suffering from cold
   vi. People with diabetes usually experience thirsty and frequent urination

7. Form two students of a certain school were interested to know the application of anaerobic respiration in their daily life. Assume you are a biology teacher; explain to these students the application of anaerobic respiration in their daily life (six points)

8. a) A woman of blood group A claims that a man of blood group B is a father of her child. A laboratory blood group test reveals that the child's blood group is O. As the biologist answer the following questions by using punnet square. Is it possible that the woman's claim is correct? Prove

   b) Variation is the possession of characteristics different from those of the parents and other off springs. In three points briefly explain the causes of this process in living organisms.

SECTION C: (30 Marks)

Answer any two (2) questions from this section.

9. Cholera is an epidemic disease which affect large number of people within a short period of time in many villages. Assume you are the only expert of biology in your village, educate your fellow villagers about that disease by writing a speech basing on the following guidelines:
   - Cause of cholera
   - Mode of transmission
   - Sign and symptoms of cholera
   - Preventive measures of cholera
   - Effects of cholera

10. Drug abuse is one of the current social problems in our country. As an educated person, educate the youths about;
    a) The meaning of drug abuse
    b) Reasons for taking drugs
    c) The effects of drug addiction

11. During the rainy season drowning become more common accident at home and school. Using four (4) points explain ways to show how you can prevent the accident.